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ORIGINAL ARTICLE

Allosterism-based simultaneous, dual anticoagulant and antiplatelet action: allosteric inhibitor targeting the glycoprotein Ibα-binding and heparin-binding site of thrombin

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Essentials

- Allosteric inhibition is a promising method for reducing bleeding risk associated with anticoagulants.
- Sulfated β-O4 lignin (SbO4L) targets exosite 2 of thrombin to induce antithrombotic effects.
- SbO4L exhibits dual anticoagulant and antiplatelet effects without increasing tail bleeding time.
- This work presents a novel molecule exploiting a novel mechanism of anticoagulation.

Summary. *Background:* Allosteric inhibition is a promising approach for developing a new group of anticoagulants with potentially reduced bleeding consequences. Recently, we designed sulfated β-O4 lignin (SbO4L) as an allosteric inhibitor that targets exosite 2 of thrombin to reduce fibrinogen cleavage through allostery and compete with glycoprotein Ibα to reduce platelet activation. *Objective:* To assess: (i) the antithrombotic potential of a novel approach of simultaneous exosite 2-dependent allosteric inhibition of thrombin and competitive inhibition of platelet activation; and (ii) the promise of SbO4L as the first-in-class antithrombotic agent. *Methods:* A combination of whole blood thromboelastography, hemostasis analysis, mouse arterial thrombosis models and mouse tail bleeding studies were used to assess antithrombotic potential. *Results and Conclusions:* SbO4L extended the clot initiation time, and reduced maximal clot strength, platelet contractile force, and the clot elastic modulus, suggesting dual anticoagulant and antiplatelet effects. These effects were comparable to those observed with enoxaparin. A dose of 1 mg of SbO4L per mouse prevented occlusion in 100% of arteries, and lower doses resulted in a proportionally reduced response. Likewise, the time to occlusion increased by ~ 70% with a 0.5-mg dose in the mouse Rose Bengal thrombosis model. Finally, tail bleeding studies demonstrated that SbO4L does not increase bleeding propensity. In comparison, a 0.3-mg dose of enoxaparin increased the bleeding time and blood volume loss. Overall, this study highlights the promise of the allosteric inhibition approach, and presents SbO4L as a novel anticoagulant with potentially reduced bleeding side effects.

Keywords: allosteric regulation; anticoagulant agents; antiplatelet agents; blood coagulation; glycoprotein Ib; thrombin.

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Introduction

Thrombotic disorders such as ischemic heart disease and venous thrombosis are major causes of mortality in humans [1]. The anticoagulants heparin and warfarin are used to treat such conditions [2–4], although both suffer from serious bleeding risks and adverse reactions. Low molecular weight heparins (LMWHs) have better safety profiles but are not completely devoid of bleeding risks [2–5]. Newer agents introduced in the clinic, including direct thrombin and factor Xa inhibitors, appear to have

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considerably improved outcomes but retain significant bleeding risks [6–8]. Likewise, antiplatelet agents, which include $P2Y_{12}$ receptor antagonists (ticlopidine, clopidogrel, and prasugrel), fibrinogen mimetics of glycoprotein (GP)IIb–IIIa receptor (tirofiban and eptifibatide), and aspirin, the inhibitor of cyclooxygenase-1 [9], are also used to treat prothrombotic conditions, but are not devoid of adverse effects. Bleeding remains a common side effect [10,11]. Additionally, many drug-specific side effects are known, such as resistance to $P2Y_{12}$ receptor antagonists [12], thrombocytopenia [13], and dosing issues in patients with renal insufficiency [14].

Although, to date, no clinically viable agent has been developed to simultaneously function as an anticoagulant (prevent fibrin formation) and antiplatelet (prevent platelet aggregation), medical conditions that require the combined use of the two types of drugs do exist [15]. For example, combination therapy has proven to be vital in the management of thrombosis in patients with prosthetic heart valves [16]. Another condition is acute arterial thrombosis, which is a common cause of myocardial infarction [15]. However, a common problem with combination therapy is the significantly increased risk of bleeding [17,18], which may arise from complete knockdown of the two key pathways of hemostasis.

Clot formation is a highly interconnected sequence of events that revolves around chemical bond cleavage and synthesis to generate polymeric fibrin, and the physical association of cells/molecules to form a platelet plug. One of these interconnections is the interaction of thrombin with GPIbα present on the platelet surface [19,20]. Detailed studies performed to date show that GPIbα binds to anion-binding exosite 2 of thrombin [21]. A specific sulfated tyrosine-rich anionic peptide sequence, ^{274}DLYsDYsYs^{279}, present on GPIbα [21,22] recognizes Arg93, Arg101, Arg126, Arg233, Lys235, Lys236, and Lys240, which are also known to be involved in binding to highly sulfated chains of heparin [23]. This cross-talk between thrombin and platelet GPIbα is important because of its key role in platelet activation and aggregation [19,20,24]. Thrombin bound to GPIbα is capable of activating protease-activated receptor-1 and GPV, which contribute to platelet aggregation, on the platelet surface [25,26]. Interestingly, GPIbα binding to thrombin induces a reduction in the catalytic efficiency of fibrinogen cleavage by thrombin [27–29].

The molecular cross-talk between fibrin formation and platelet activation/aggregation presents a novel avenue for the development of new antithrombotics [24]. Our earlier work demonstrated that sulfated molecules that target exosite 2 allosterically inhibit thrombin and show an anticoagulant effect [30–32]. If these sulfated molecules also compete with GPIbα for binding to exosite 2, then they will show an antiplatelet effect. In support of this hypothesis, we had earlier designed a molecule called sulfated β-O4 lignin (SbO4L) (Fig. 1) that competed with GPIbα for binding to thrombin and reduced aggregation in platelet-rich plasma [33]. More importantly, SbO4L allosterically inhibited thrombin's catalytic activity with nanomolar potency (IC_{50} ~ 14 nM) and caused a prolongation of human plasma clotting times in activated partial thromboplastin time and prothrombin time assays. Interestingly, mutagenesis-based experiments had shown that SbO4L binds to Arg233, Lys235 and Lys236 on thrombin, which define the site of binding for both GPIbα and heparin. However, unlike heparin, SbO4L does not require antithrombin to inhibit thrombin, and targets thrombin directly. Thus, SbO4L represents a new class of interesting molecules that show dual anticoagulant and antiplatelet activities, which rely on allostery and could potentially be useful in specialized thrombotic events.

In this article, we report on the efficacy of SbO4L as a prototypic, first-in-class, dual-action antithrombotic in preventing *in vitro* and *in vivo* clot formation. Our study shows that SbO4L effectively reduces the rate and extent of fibrin formation, while also reducing platelet contractile force and the clot elastic modulus. *In vivo*, SbO4L prevents arterial occlusion, while not enhancing tail bleeding propensity in mice. Thus, SbO4L shows considerable promise as an allosteric anticoagulant/antiplatelet agent that targets the cross-talk between thrombin and GPIbα, highlighting the value of studying novel mechanisms as an avenue for obtaining new antithrombotic agents with reduced bleeding side effects.

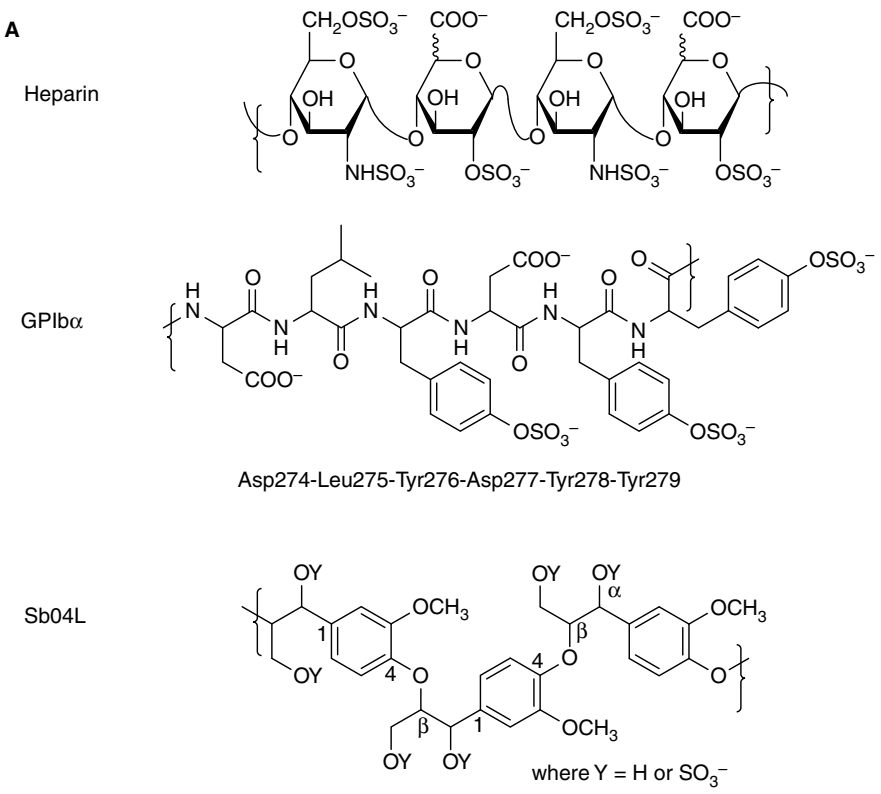
Materials and methods

Proteins and chemicals

Human and murine thrombins were obtained from Haematologic Technologies (Essex Junction, VT, USA). Protamine was obtained from Sigma-Aldrich (St. Louis, MO, USA). Spectrozyme TH was obtained from Sekisui Diagnostics (Stamford, CT, USA). The Thromboelastograph Coagulation Analyzer 5000 and its supplies were obtained from Haemoscope Corporation (Niles, IL, USA).

Fig. 1. Structural basis for a novel mechanism of dual anticoagulant and antiplatelet action. (A) Structures of the common sequence of heparin, the sulfated tyrosine-containing peptide sequence in glycoprotein (GP)Ibα, and the common sequence of fully synthetic sulfated β-O4 lignin (SbO4L). (B) A cartoon representation of the effect of heparin and GPIbα following binding to exosite 2 of thrombin. Heparin induces minimal allosteric changes in the catalytic site, which do not affect the hydrolysis of substrates. In contrast, GPIbα induces some allosteric inhibition, but contributes more to the activation of platelets. (C) A cartoon representation of the effect of SbO4L binding to exosite 2 of thrombin. Significant allosteric changes in the catalytic site of thrombin reduce cleavage of substrate, and there is simultaneous competition with GPIbα and reduction in the activation of platelets. These effects generate the anticoagulant and antiplatelet functions of SbO4L.

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Inhibition of murine thrombin

SbO4L inhibition of murine thrombin was studied with a substrate hydrolysis assay in the manner described previously for human thrombin [30–33]. Briefly, 5 μL of either water or SbO4L at 2.3 ng mL^{-1} to 2.3 mg mL^{-1} was diluted with 185 μL of 20 mM Tris-HCl buffer (pH 7.4), containing 100 mM NaCl, 2.5 mM $CaCl_2$ and 0.1% PEG-8000, in a 96-well polystyrene microplate at 37 °C. Five microliters of murine thrombin was then added (6 nM final concentration), and the solution was incubated for 10 min; 5 μL of 5 mM Spectrozyme TH was then added. The residual activity of thrombin was determined by monitoring the absorbance at 405 nm. The SbO4L concentration that results in 50% of inhibition of thrombin (IC_{50}), the Hill slope (HS) and maximal percentage residual thrombin activity (Y_M) were calculated with logistic equation 1, in which Y is the percentage residual thrombin activity at each concentration of SbO4L, and Y_0 is the percentage residual thrombin activity in the absence of SbO4L.

$$Y = Y_0 + \frac{Y_M - Y_0}{1 + 10^{((\log[Sb04L]_0 - \log IC_{50}) \times HS)}}$$

Reversal of SbO4L activity by protamine

Protamine (0–14.8 mg mL^{-1}) was added to a solution of 6 nM thrombin and 0.58 μg mL^{-1} SbO4L in 20 mM Tris-HCl buffer (pH 7.4), containing 100 mM NaCl, 2.5 mM $CaCl_2$ and 0.1% PEG-8000, at 37 °C. Immediately thereafter, the residual thrombin activity was measured according to Spectrozyme TH hydrolysis, as described above. Control experiments with protamine alone showed that the activity of thrombin was unaffected. The increase in thrombin activity at varying protamine concentrations could be fitted by a modified form of logistic equation 1 to calculate the protamine concentration that restores thrombin activity by 50% (RC_{50}).

Human whole blood thromboelastography (TEG) analysis of clot formation

TEG assays were performed with human whole blood in the presence of SbO4L, as described previously [34]. Briefly, aliquots of whole blood were incubated with an aqueous solution of SbO4L (10 μL) for 5 min to yield final concentrations of 0–152 μg mL^{-1}. TEG assays were carried by transferring 20 μL of 200 mM $CaCl_2$ into the TEG cup, incubating at 37 °C, and transferring 340 μL of whole blood into the cup to start the assay. The TEG cup oscillated through an angle of 4°45′ at 0.1 Hz until all data were collected; these included the time to initiation of clot (reaction time [R]), the kinetics of clot formation (angle [α]), the maximal stiffness (maximal amplitude [MA]), and the elasticity (shear elastic modulus [G]) of the clot formed, as discussed previously [34–36]. All assays were run in duplicate.

Human whole blood and platelet-rich plasma (PRP) hemostatic analysis

Analysis of platelet function and clot structure was performed with the Hemostasis Analysis System (HAS) from Hemodyne (Richmond, VA, USA), as reported previously [34]. A mixture of 700 μL of citrated whole blood or PRP (platelet count adjusted to 200 000 per μL) and 10 μL of SbO4L (or ddH_2O as control) to give 0–74 μg mL^{-1} was coincubated at room temperature for 5 min, and then placed in a disposable cup. To initiate clotting, 50 μL of 150 mM $CaCl_2$ was added, and the HAS was used to measure the platelet contractile force (PCF), the platelet force onset time, and the clot elastic modulus (CEM). All assays were performed in duplicate.

In vivo *$FeCl_3$ carotid artery thrombosis studies*

Procedures involving mice were approved by the Institutional Animal Care and Use Committee of Vanderbilt University, and have been previously reported [37,38]. For the $FeCl_3$ carotid artery thrombosis model, wild-type C57Bl/6 mice (n = 4–8; weight, 22–27 g) were anesthetized with 50 mg kg^{-1} intraperitoneal pentobarbital. SbO4L (0, 100, 300, 500 or 1000 μg in 100 μL of phosphate-buffered saline [PBS]) was infused into the right internal jugular vein. Five minutes after infusion, the right common carotid artery was exposed and fitted with a Doppler flow probe (Model 0.5 VB; Transonic System, Ithaca, NY, USA). Thrombus formation was induced by applying two 1 × 1.5-mm filter papers (GB003; Schleicher & Schuell, Keene, NH, USA) saturated with $FeCl_3$ (3.5% w/v solution) to opposite sides of the artery for 3 min. After washing of the site of injury with PBS, blood flow was monitored for 30 min. Mice were killed with a pentobarbital overdose after conclusion of the experiment and under anesthesia.

In vivo *Rose Bengal laser injury arterial thrombosis studies*

In vivo testing with Rose Bengal thrombosis model studies was performed as described previously [38]. Briefly, wild-type C57Bl/6 mice (n = 3 per group; weight, 22–27 g) were anesthetized as described above, and 100 μL of PBS with or without 500 μg of SbO4L was infused into the right internal jugular vein. Five minutes after infusion, Rose Bengal (75 mg kg^{-1}) was infused through the internal jugular vein, and the carotid artery was illuminated with a 1.5-mW 540-nm laser (Melles Griot, Carlsbad, CA, USA) positioned 6 cm from the artery. Flow was monitored for 120 min. Mice were killed with a pentobarbital overdose after conclusion of the experiment and under anesthesia.

Tail bleeding time studies

Wild-type ICR mice ($n \geq 5$; weight, 30–45 g) were anesthetized with an intraperitoneal injection of pentobarbital

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(50 mg kg^{-1}), and tail bleeding was performed as described previously [39]. Briefly, following anesthesia, the mice were placed on a 37 °C heating pad. Either LMWH (0.1–0.5 mg), SbO4L (1 mg) or control (PBS), prepared in 50 μL of PBS, was injected into the lateral tail vein with a syringe. After 2 min, the tail tip (~ 4–5 mm) was transected with a scalpel and immediately immersed in a 1.7-mL microtube filled with 1.2 mL of PBS placed in a 37 °C heating block. The mice were observed for 15 min. The total bleed time (defined as the sum of all bleeding events occurring within 15 min) and the weight of blood lost were recorded for each mouse.

Results

SbO4L is an inhibitor of thrombin hydrolysis of its small and large substrates

Among the animal models of thrombosis, mouse models have been developed the most [39]. There are small structural differences between mouse and human thrombins, especially within the sodium-binding site (Protein Data Bank ID:2OCV), which is known to allosterically regulate the catalytic triad [40,41]. It has been reported that the sodium-binding site is also energetically coupled to exosites 1 and 2 [42], exosite 2 being the site of heparin and GPIbα binding. This raises the question of whether SbO4L inhibits murine thrombin. We measured SbO4L inhibition of murine thrombin by using the Spectrozyme TH hydrolysis assay (Fig. 2A), as described previously [33], and measured an IC_{50} of 0.17 μg mL^{-1}, which is essentially identical to the potency measured against the human version.

To also assess whether SbO4L inhibits thrombin cleavage of fibrinogen, we utilized a transmittance assay that quantifies the formation of fibrin in buffer. Using this assay, we found that SbO4L inhibited thrombin cleavage of fibrinogen with an IC_{50} of 0.19 μg mL^{-1} (Fig. S1), which is comparable to that measured with a small-molecule chromogenic substrate, further confirming its possible value as an anticoagulant.

SbO4L inhibition of thrombin can be reversed by protamine

Protamine, an arginine-rich highly cationic protein mixture, is used to neutralize the effects of highly sulfated heparin both *in vitro* and *in vivo*. It is a Food and Drug Administration (FDA)-approved antidote for treating heparin overdosing, and offers a major advantage over heparin-based anticoagulation therapy. Because SbO4L is also a sulfated agent, we reasoned that protamine may be able to neutralize its thrombin inhibition potential. To test this, we studied the reversibility of SbO4L inhibition of thrombin by protamine by using the Spectrozyme TH hydrolysis assay. Figure 2B shows the gain in thrombin catalytic activity in the presence of increasing levels of

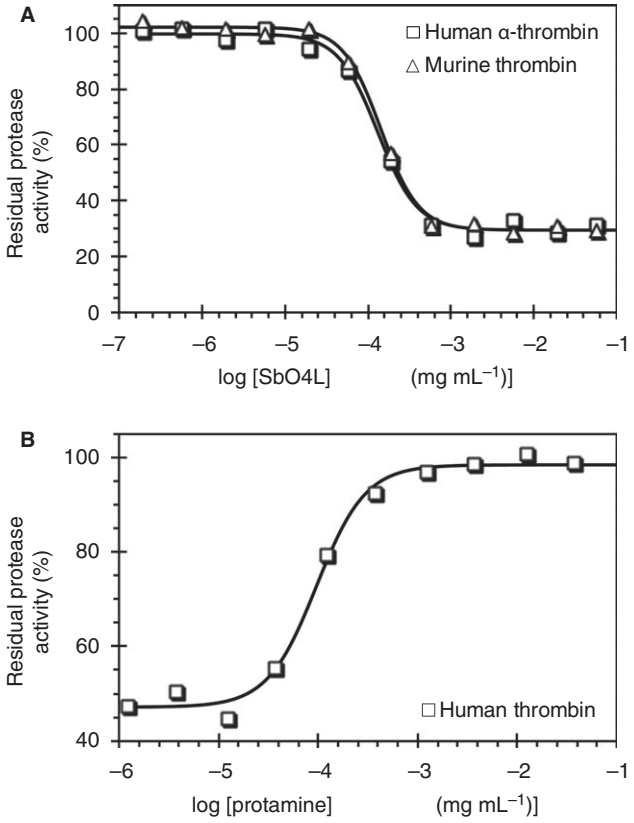


Fig. 2. A comparison of direct inhibition of human and murine thrombins by sulfated β-O4 lignin (SbO4L) (A) and recovery of human thrombin activity in the presence of protamine (B). The concentration of residual thrombin activity at pH 7.4 and 37 °C was measured by chromogenic substrate hydrolysis assay, as described previously [33]. Solid lines represent sigmoidal dose–response fits (Eqn 1) to the data to obtain IC_{50} (A) or RC_{50} (B), the Hill slope, and maximal inhibition or recovery efficacy. The definitions of these terms are given in Materials and methods.

protamine. The profile essentially mirrors the inhibition profile, and the recovery was instantaneous. The RC_{50} was found to be ~ 0.1 μg mL^{-1}, which is comparable to the IC_{50} for SbO4L inhibition of thrombin. Also, protamine reversed SbO4L inhibition of thrombin by 100%, which can be expected to have significant advantages.

SbO4L shows good anticoagulant potential in human whole blood thromboelastography

To evaluate SbO4L as an anticoagulant in whole blood, we employed TEG, which attempts to simulate clotting under low-shear conditions. TEG has been approved by the FDA for the diagnosis and management of coagulation disorders [43]. TEG is also quite often used to monitor LMWH therapy, and is especially useful for assessing the anticoagulant state during surgery and coagulopathy [35,36]. TEG can easily monitor the time to initiation and rate of clot formation; however, it is also useful for assessing the nature of physical forces within a clot. A good anticoagulant should both slow clot formation

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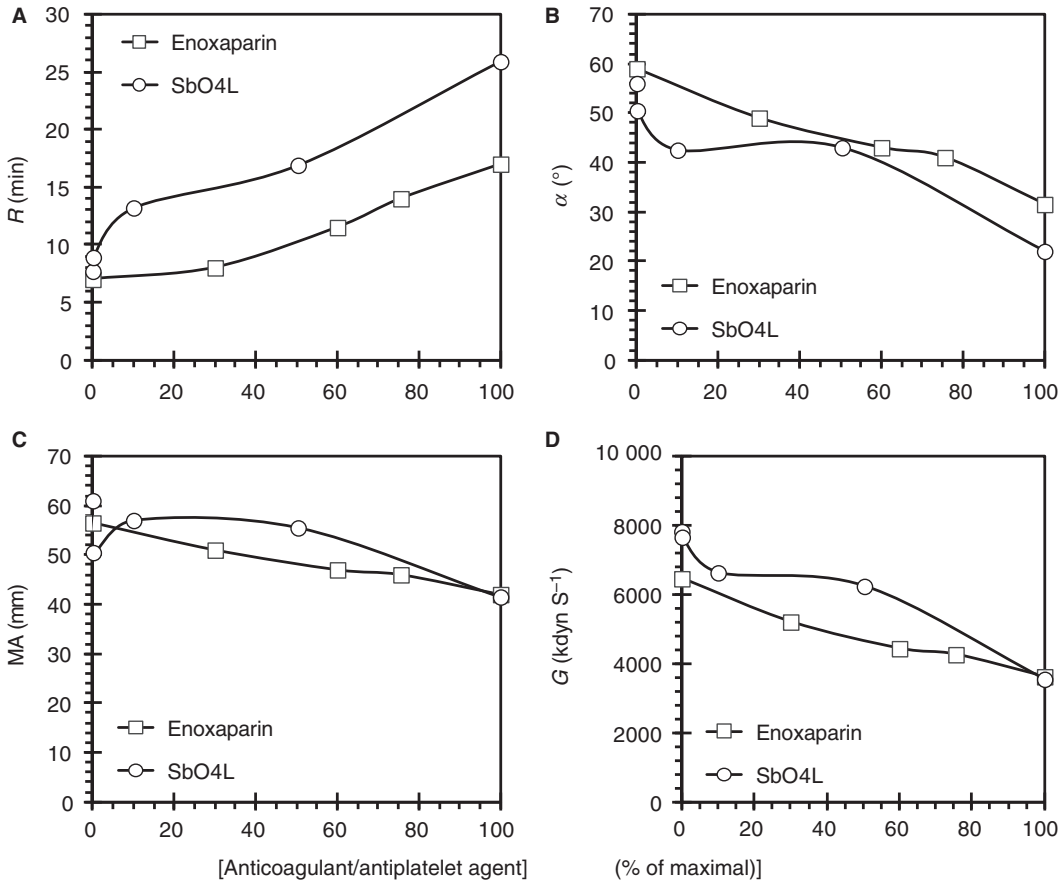


Fig. 3. A comparison of the effects of sulfated β-O4 lignin (SbO4L) and enoxaparin on clot formation in human whole blood determined with thromboelastography. (A) and (B) show the changes in reaction time (R) and angle (α), respectively; (C) and (D) show the changes in maximal amplitude (MA) and shear elastic modulus (G), respectively. These parameters are automatically calculated by the instrument manufacturer's algorithm following an experiment, as described previously [34–36]. SbO4L and enoxaparin concentrations are plotted as percentage of the maximum used.

and reduce the intermolecular forces within the clot. Both of these effects decrease the extent and integrity of the clot, which are quantified by the use of TEG parameters, including MA, G, R, and α [34–36].

Figure 3 shows the changes in R, α, MA and G as a function of the level of SbO4L or enoxaparin, a representative LMWH. We chose enoxaparin as a control because it is similar in homogeneity, size and charge to SbO4L. As SbO4L and enoxaparin are active at different concentrations, the effect of the TEG parameters is displayed as a percentage of the maximal concentration of each agent used in the experiment. As the level of SbO4L increased from 0% to 100%, R increased from 7.7 min to 25.9 min, whereas α decreased from 56° for normal blood to 22°, indicating that the kinetics of fibrin polymerization and network formation are significantly depressed in the presence of SbO4L (Fig. 3A,B). Essentially identical profiles for R and α were measured for enoxaparin at the respective percentage levels used in the experiment. Likewise, the extent of fibrin polymerization and integrity of the clot, i.e., MA and G, respectively, decreased with increasing levels of SbO4L (Fig. 3C,D). As expected, enoxaparin results were found to be similar to the SbO4L results. Thus, overall, both agents caused very similar changes in TEG parameters; however, their dosage levels were different. SbO4L was active in the range 5–150 μg mL^{-1}, whereas enoxaparin was active in the range 1–5 μg mL^{-1}.

SbO4L shows good antiplatelet activity in human whole blood hemostatic analysis

To further assess the antithrombotic potential of SbO4L, we utilized the HAS, which evaluates the platelet contribution to clot formation [44–46]. This technique provides information on contractile forces between platelets adhering to surfaces that restrict the relative movement of two cups. The force measured, called the PCF, depends on several factors, including the number and status of platelets and the presence of inhibitors that affect platelet activation [44–47]. The HAS also assesses the internal microstructure of a growing thrombus in the form of the CEM, which is the ratio of stress induced by platelets to strain arising from the change in clot thickness. The CEM also depends on several factors, including fibrino-

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gen levels, the activity and level of FXIII, and the thrombin formation rate [44–48]. It has been suggested that changes in the PCF and CEM correlate with susceptibility to bleeding and/or thrombotic tendency [44,49,50].

The effects of SbO4L on the PCF and CEM are shown in Fig. 4 and Table 1. Both the PCF and the CEM are affected by SbO4L in a dose-dependent manner. As the concentration of SbO4L increases from 0 μg mL^{-1} to 74 μg mL^{-1}, the PCF and CEM decrease from 8.0 kdyn to 0.7 kdyn and 18.3 kdyn cm^{-2} to 0.6 kdyn cm^{-2}, respectively. These results parallel those measured with TEG. When comparisons were made with enoxaparin, essentially identical results were obtained, except for the difference in the concentration of enoxaparin necessary to prevent clotting. Whereas a reduction in the PCF value of < 1.0 kdyn was achieved at 74 μg mL^{-1} (or 6.2 μM) for SbO4L, it was achieved at 2.0 μg mL^{-1} (0.4 μM) for enoxaparin. Comparable results were also obtained at these doses when PRP was used (platelet count of ~ 200 000 per μL) instead of human whole blood (Fig. S2). These results suggest that SbO4L-mediated targeting of the GPIbα site of thrombin exosite 2 inhibits platelet activation, resulting in significant reductions in the PCF and CEM.

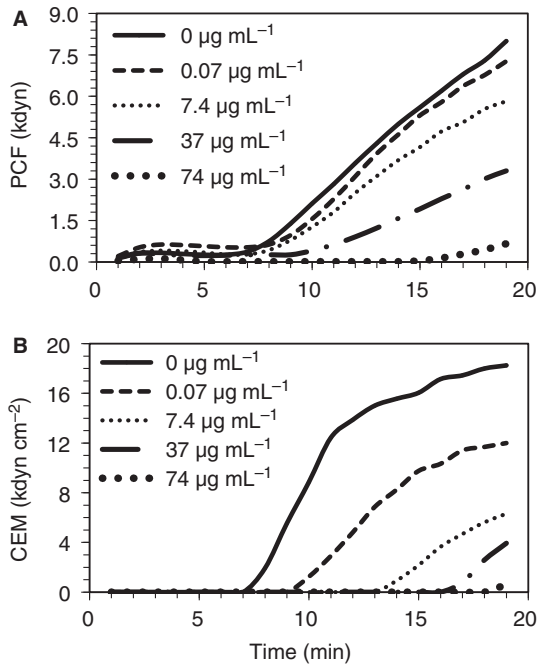


Fig. 4. The effect of sulfated β-O4 lignin (SbO4L) on platelet function in human whole blood determined with the Hemostasis Analysis System. (A) and (B) show the changes in platelet contractile force (PCF) and clot elastic modulus (CEM), respectively, with time at varying concentrations of SbO4L (0–74 μg mL^{-1}). See also Table 1 for quantitative variation of PCF and CEM at different SbO4L concentrations.

SbO4L shows good antithrombotic efficacy in mouse arterial thrombosis models

To assess whether the anticoagulant and antiplatelet activities of SbO4L translate into *in vivo* antithrombotic activity, we utilized two well-characterized mouse arterial thrombosis models, i.e. the $FeCl_3$-induced [39] and the Rose Bengal-induced [39,51] artery injury models. Both models have been extensively utilized for studying anticoagulants [39,52], although neither model exactly simulates the prothrombotic state that humans typically experience. In fact, $FeCl_3$ is considered to produce a severe injury, in comparison with a prothrombotic state in humans, and is typically accompanied by desquamation of vascular endothelium [53]. This implies that an agent that suppresses arterial occlusion following $FeCl_3$ injury is likely to show good antithrombotic potential in higher animals. Finally, detailed studies of dose-dependent reductions in thrombosis with the $FeCl_3$-induced arterial injury model have been published for enoxaparin [54,55] and full-length heparin [37].

Exposure of the carotid artery of wild-type mice to a 3.5% solution of $FeCl_3$ resulted in the formation of an

Table 1 Hemostasis Analysis System (HAS) parameters for sulfated β-O4 lignin (SbO4L) anticoagulation in comparison with enoxaparin

		Hemostasis analysis*			
	Concentration (μg mL^{-1})	Concentration† (μM)	FOT* (min)	PCF* (kdyn)	CEM* (kdyn cm^{-2})
SbO4L	0	0	6.3	8.0	18.3
	0.07	0.006	5.1	7.3	12.0
	7.4	0.617	6.6	5.8	6.3
	37	3.080	11.3	3.3	3.9
	74	6.170	13.6	0.7	0.6
Enoxaparin	0	0	6.2	7.6	21.6
	0.7	0.156	5.2	5.3	15.1
	1.0	0.222	9.1	3.6	12.7
	1.6	0.356	11.0	2.8	8.5
	2.0	0.444	12.5	0.9	2.9

CEM, clot elastic modulus; FOT, force onset time; PCF, platelet contractile force. *Analysis was performed with the HAS on human whole blood as described. †Calculated by using average molecular masses of 12 000 Da for SbO4L and 4500 Da for enoxaparin.

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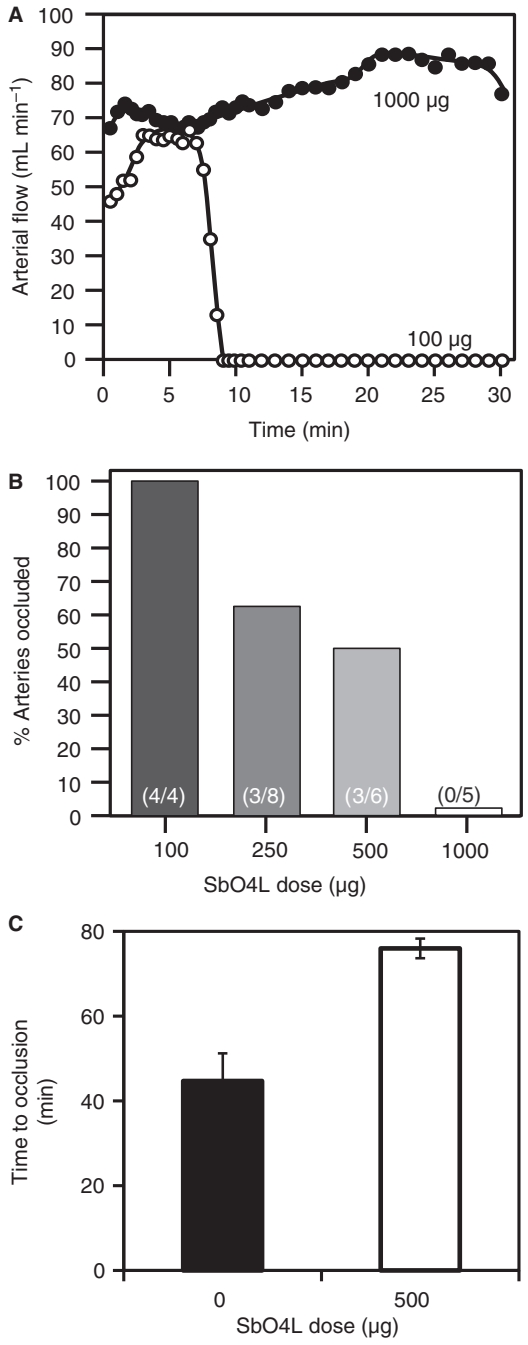


Fig. 5. *In vivo* studies on the effect of sulfated β-O4 lignin (SbO4L) in two mouse models of arterial thrombosis. (A) Time dependence of the formation of occlusive platelet-rich thrombus in the carotid artery of mice resulting from application of a 3.5% $FeCl_3$ solution with two doses of SbO4L, i.e. 100 µg and 1000 µg. (B) Dose-dependent decrease in occlusion as the levels of SbO4L increase from 100 µg to 1000 µg per mouse. The numbers in parentheses indicate the fraction of mice that showed complete and sustained occlusion for 15 min. (C) Results obtained in the Rose Bengal laser injury-induced arterial thrombosis model with a dose of 500 µg of SbO4L per mouse (or phosphate-buffered saline injection).

occlusive platelet-rich thrombus within 15 min. When SbO4L was injected ~ 10 min before the application of the $FeCl_3$ patch, a distinct reduction in the extent of thrombus formation was observed (Fig. 5). At 100 µg per mouse, no significant effect was observed, probably because thrombin was not saturated, but at 250 µg or 500 µg, SbO4L prevented occlusion in approximately 35–50% of arteries studied. In contrast, administration of 1 mg of SbO4L prevented occlusion in 100% of arteries studied (Fig. 5). In comparison, literature reports indicate that a comparable ~ 1-mg dose of enoxaparin results in an approximately 50% reduction in thrombosis in identical $FeCl_3$-induced thrombosis models [54,55].

The effects of SbO4L were also tested in the Rose Bengal–laser injury model of arterial thrombosis, in which the primary mechanism for generating a vessel injury is through the local production of free radicals [39,51]. A focused beam of light converts the Rose Bengal dye to free radicals, which initiate the formation of a platelet-rich thrombus. Control experiments indicated that vessel occlusion was completed in approximately 45 ± 7 min for wild-type mice. In contrast, vessel occlusion occurred in 76 ± 21 min in mice dosed with 500 µg of SbO4L ~ 10 min before laser injury (Fig. 5C). This represents a significant prolongation in clotting, suggesting a potent antithrombotic effect of SbO4L. An important point to note here is that SbO4L appears to be more potent than

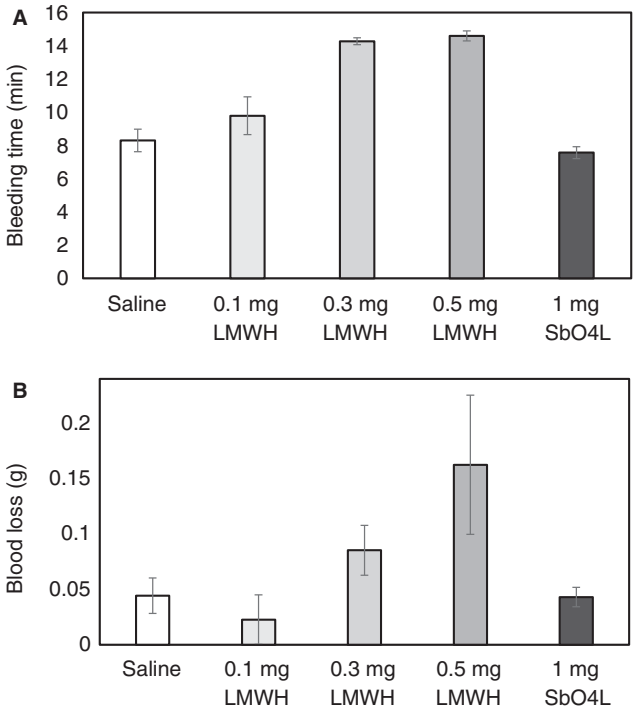


Fig. 6. A comparison of mouse tail bleeding following intravenous injection of either 1 mg of sulfated β-O4 lignin (SbO4L), with 0.1, 0.3 or 0.5 mg of enoxaparin, or phosphate-buffered saline. Two parameters were followed: bleeding time (A) and total blood loss (B). Error bars represent ± 1 standard error of the mean ($N = 5$). SbO4L was found to have a tendency to induce bleeding that was less than that of enoxaparin and not significantly different from that of vehicle. LMWH, low molecular weight heparin.

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enoxaparin in *in vivo* antithrombotic assays, whereas the reverse is true for *in vitro* assays. Although the basis for the better performance of SbO4L *in vivo* is not clear, it is likely to be synergy brought about the unique dual anticoagulant–antiplatelet function.

SbO4L does not enhance tail bleeding

To assess the propensity of SbO4L to enhance bleeding, the murine tail bleeding experiment, which has been utilized in a large number of studies, was performed [39]. Figure 6 shows the total bleeding time and the total blood loss for 1 mg of SbO4L, 0.1 mg, 0.3 mg and 0.5 mg of enoxaparin and PBS injected ~ 10 min before the transection of the tail. For SbO4L, the tail bleeding time was found to be 7.6 ± 0.4 min, which was essentially the same as that noted for PBS (8.3 ± 0.7 min). Likewise, the total blood lost following tail clipping for SbO4L was 0.043 ± 0.008 mg, which was also identical to that noted for PBS (0.044 ± 0.016 mg). In comparison, a three-fold lower dose of enoxaparin (0.3 mg per mouse) resulted in a bleeding time of 14.3 ± 0.2 min and blood loss of 0.085 ± 0.022 mg. These represent an approximately 1.7-fold higher bleeding time and approximately 1.9-fold more blood loss for enoxaparin than for SbO4L. Similar results have been reported previously [54,55]. To identify which dose of enoxaparin would produce results similar to SbO4L, we studied 0.1-mg, 0.2-mg, 0.4-mg and 0.5-mg doses. In all cases, enoxaparin resulted in a higher bleeding time than SbO4L, and the same or higher blood loss (Fig. 6; data shown only for 0.1-mg and 0.5-mg doses). Thus, SbO4L appears to result in a reduced bleeding tendency in comparison with enoxaparin.

Discussion

This study brings forth two primary ideas: (i) direct, allosteric inhibition of thrombin can generate sufficient antithrombotic potential, and may also provide an advantage in terms of reducing bleeding risk; and (ii) SbO4L is a novel prototypic, first-in-class molecule that simultaneously shows good anticoagulant and antiplatelet properties by targeting the GPIbα/heparin-binding site on thrombin. Except for hirudin and bivalirudin, all direct inhibitors used in the clinic today are active site inhibitors. Even hirudin and bivalirudin bind in the active site, while also engaging exosite 1 of thrombin. Thus, SbO4L represents an attempt to establish a novel approach of direct, allosteric inhibition in developing new antithrombotics.

Secondary concepts highlighted include: (i) targeting the heparin-binding and GPIbα-binding site of thrombin; and (ii) dual anticoagulant and antiplatelet function in the same molecule. Although unfractionated heparin binds in exosite 2 of thrombin, on its own it does not induce inhibition of thrombin, and it has to rely on antithrombin, whereby it bridges thrombin and antithrombin to form a covalent complex. This implies that structural features of SbO4L, although mimicking the highly sulfated feature of heparin, are important for inducing an allosteric conformational change. We have shown previously that SbO4L recognizes hydrophobic subdomains present within exosite 2, inducing allosteric structural changes [33,56]. SbO4L targets overlapping binding sites of heparin and GPIbα, which advantageously elicits the dual anticoagulant and antiplatelet function. No therapeutic agent has been developed to date that targets this overlapping binding site, which suggests that it may be possible to develop better analogs through structure-based drug discovery.

A common concern with dual anticoagulant and antiplatelet agents is the possibility of elevated bleeding. This study shows that this may not necessarily be true, and additional studies should be performed to further establish the concept of reduced bleeding by targeting the dual heparin-binding and GPIbα-binding site of thrombin. The exact reason why SbO4L shows a lower tendency to cause bleeding in tail bleeding studies is yet to be deciphered. We have pursued allosterism as a mechanism to discover new anticoagulants because of its potential in enhancing the specificity of action. Allosteric binding sites on proteins are thought to be more structurally diverse, especially for related proteins, which should reduce the chances of cross-reactivity [57]. In fact, SbO4L is highly selective in targeting thrombin, and does not affect the closely related FXa [33]. Another reason why SbO4L does not have a higher tendency to cause bleeding could be its reduced charge density in comparison with heparins. Whereas heparins possess an average of 2.7 negative charges per repeating unit [58], SbO4L possesses only 1.7 sulfate groups per monomer [33]. Higher charge density tends to induce binding to many blood proteins and cells, which may enhance bleeding [59].

This study presents SbO4L as the first-in-class allosteric agent that simultaneously induces anticoagulation and antiplatelet actions. SbO4L is not the final therapeutic agent that is ready for clinical studies. However, it offers a number of advantages. It was possible to reverse its antithrombotic activity *in vitro*. Second, although not presented here, SbO4L is a fully synthetic molecule that can be prepared in two simple steps from common raw materials in fairly high yields [33]. SbO4L is also much better structurally defined than the heterogeneous heparins used in the clinic. SbO4L itself has a fairly uniform composition (Fig. 1) that can be monitored with ease with common analytic techniques, which implies that advanced versions of this molecule could be more easily assessed for quality control. Overall, direct, selective, allosteric inhibition of thrombin through SbO4L appears to be a promising approach that is worthy of advanced antithrombotic and toxicologic studies.

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Addendum

A. Y. Mehta performed SbO4L synthesis and biochemical experiments, helped with TEG and HAS experiments, and prepared the initial draft of the manuscript. B. M. Mohammed, E. J. Martin, and D. F. Brophy helped with TEG, HAS and tail bleeding studies. D. Gailani supervised arterial thrombosis studies. U. R. Desai supervised the entire study, and finalized the paper.

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Disclosure of Conflict of Interests

A. Y. Mehta and U. R. Desai have a patent for Sulfated Beta-O4 Low Molecular Weight Lignins pending. D. Gailani reports grants from the National Heart, Lung and Blood Institute, during the conduct of the study; nonfinancial support from Aronora, Bayer, Bristol-Myers Squibb, and Dyax; and personal fees from Isis, Novartis, Ono Phmaraceutical, Instrument Laboratory, and Merck, outside the submitted work. In addition, D. Gailani has a patent for Anti-Factor XI Antibodies with royalties paid. The other authors state that they have no conflict of interest.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Fig. S1. Direct inhibition of human thrombin by SbO4L with fibrinogen as substrate.
Fig. S2. The effect of SbO4L on platelet function in human PRP determined with the Hemostasis Analysis System (HAS).

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